WORKED SOLUTION

STEP III 2018, Question 1

Pure • Quadratic Roots and Inequalities

STEP III 2018, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s3-q1

The key idea

Vieta's formulas turn both symmetric expressions in u and v into simple functions of the coefficients α and β. Once one expression is fixed, α can be eliminated, and the other expression becomes a function of β alone: exactly one of the two functions sketched in part (i). The condition that u and v are real (discriminant ≥ 0) then tells us which stretch of the curve we are allowed to use, and the sketch does the rest.

Solution

Part (i)

The curve $y = \mathrm{f}(\beta)$. Differentiating,

$$\mathrm{f}'(\beta) = 1 + \frac{1}{\beta^2} + \frac{2}{\beta^3} = \frac{\beta^3 + \beta + 2}{\beta^3} = \frac{(\beta+1)(\beta^2 - \beta + 2)}{\beta^3},$$

where we spotted the root $\beta = -1$ of the cubic. The quadratic factor has discriminant $1 - 8 < 0$, so it is always positive. Hence the only stationary point is at $\beta = -1$, where

$$\mathrm{f}(-1) = -1 + 1 - 1 = -1.$$

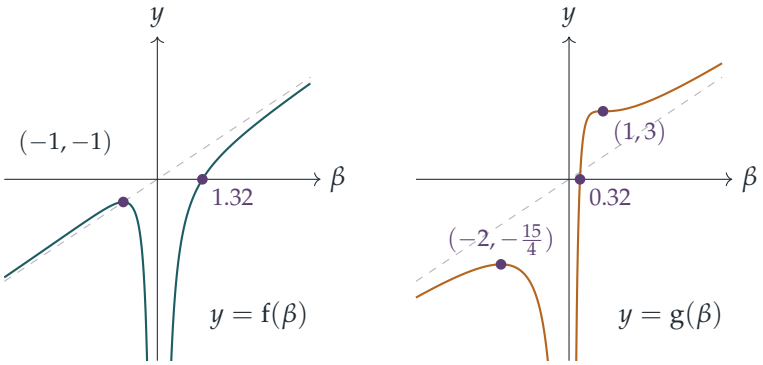
Signs: for $-1 < \beta < 0$ the numerator is positive and $\beta^3 < 0$, so $\mathrm{f}' < 0$; for $\beta < -1$ both are negative, so $\mathrm{f}' > 0$. So $(-1, -1)$ is a *maximum*, and $\mathrm{f}(\beta) \leq -1$ for every $\beta < 0$. For $\beta > 0$ everything is positive, so f is increasing there.

Behaviour at the edges: as $\beta \to 0$ from either side the term $-1/\beta^2$ dominates, so $\mathrm{f} \to -\infty$. As $\beta \to \pm\infty$, $\mathrm{f}(\beta) - \beta \to 0$, so $y = \beta$ is an oblique asymptote. The curve crosses the axis once, where $\beta^3 - \beta - 1 = 0$, at $\beta \approx 1.32$.

The curve $y = \mathrm{g}(\beta)$. Similarly,

$$\mathrm{g}'(\beta) = 1 - \frac{3}{\beta^2} + \frac{2}{\beta^3} = \frac{\beta^3 - 3\beta + 2}{\beta^3} = \frac{(\beta-1)^2(\beta+2)}{\beta^3}.$$

The stationary points are at $\beta = -2$, with $g(-2) = -2 - \frac{3}{2} - \frac{1}{4} = -\frac{15}{4}$, and at $\beta = 1$, with $g(1) = 3$. For $-2 < \beta < 0$ we have $g' < 0$, and for $\beta < -2$ we have $g' > 0$, so $(-2, -\frac{15}{4})$ is a maximum and $g(\beta) \leq -\frac{15}{4}$ for all $\beta < 0$. For $\beta > 0$, $g' \geq 0$ with equality only at $\beta = 1$: the derivative does not change sign there, so $(1, 3)$ is a stationary point of inflection and g is increasing on $\beta > 0$. Again $g \to -\infty$ as $\beta \to 0$, $y = \beta$ is an asymptote, and there is one zero, at $\beta \approx 0.32$ (the root of $\beta^3 + 3\beta - 1 = 0$).



The dashed line is the asymptote $y = \beta$.

Part (ii)

By Vieta's formulas, $u + v = -\alpha$ and $uv = \beta$ (and $uv \neq 0$ since $\beta \neq 0$). So

$$u + v + \frac{1}{uv} = -\alpha + \frac{1}{\beta}, \qquad \frac{1}{u} + \frac{1}{v} + uv = \frac{u+v}{uv} + uv = \beta - \frac{\alpha}{\beta}.$$

Part (iii)

The condition $-\alpha + \frac{1}{\beta} = -1$ gives $\alpha = 1 + \frac{1}{\beta}$. Substituting into the second expression,

$$\frac{1}{u} + \frac{1}{v} + uv = \beta - \frac{1}{\beta}\left(1 + \frac{1}{\beta}\right) = \beta - \frac{1}{\beta} - \frac{1}{\beta^2} = f(\beta).$$

So we must show $f(\beta) \leq -1$ for every *allowed* β.

Ask yourself The curve $y = f(\beta)$ goes off to $+\infty$, so the inequality can't hold for all β. What stops β from being large?

The reality of u and v. The roots are real exactly when $\alpha^2 \geq 4\beta$, that is,

$$\left(1 + \frac{1}{\beta}\right)^2 \geq 4\beta \iff (\beta+1)^2 \geq 4\beta^3 \iff 4\beta^3 - \beta^2 - 2\beta - 1 \leq 0,$$

where we multiplied by $\beta^2 > 0$. Now $\beta = 1$ is a root of the cubic, and

$$4\beta^3 - \beta^2 - 2\beta - 1 = (\beta - 1)(4\beta^2 + 3\beta + 1).$$

The quadratic factor has discriminant $9 - 16 < 0$, so it is always positive. Hence the roots are real if and only if $\beta \leq 1$ (with $\beta \neq 0$).

- If $\beta < 0$: from part (i), $f(\beta) \leq f(-1) = -1$.
- If $0 < \beta \leq 1$: f is increasing on $\beta > 0$, so $f(\beta) \leq f(1) = 1 - 1 - 1 = -1$.

In every case $\frac{1}{u} + \frac{1}{v} + uv \leq -1$, as required. Equality occurs at $\beta = -1$ ($\alpha = 0$, roots ± 1) and at $\beta = 1$ ($\alpha = 2$, double root -1).

Part (iv)

Now $-\alpha + \frac{1}{\beta} = 3$, so $\alpha = \frac{1}{\beta} - 3$ and

$$\frac{1}{u} + \frac{1}{v} + uv = \beta - \frac{1}{\beta}\left(\frac{1}{\beta} - 3\right) = \beta + \frac{3}{\beta} - \frac{1}{\beta^2} = g(\beta).$$

The roots are real when $\alpha^2 \geq 4\beta$:

$$\left(\frac{1}{\beta} - 3\right)^2 \geq 4\beta \iff (1 - 3\beta)^2 \geq 4\beta^3 \iff 4\beta^3 - 9\beta^2 + 6\beta - 1 \leq 0.$$

Again $\beta = 1$ is a root, and this time it is a double root:

$$4\beta^3 - 9\beta^2 + 6\beta - 1 = (\beta - 1)(4\beta^2 - 5\beta + 1) = (\beta - 1)^2(4\beta - 1).$$

Since $(\beta - 1)^2 \geq 0$, the cubic is ≤ 0 exactly when $\beta \leq \frac{1}{4}$ *or* $\beta = 1$. So the allowed values are $\beta < 0$, $0 < \beta \leq \frac{1}{4}$, and the isolated value $\beta = 1$.

> **Ask yourself** Is the greatest value simply $g(\frac{1}{4})$, the right-hand end of the interval?

Not quite: don't lose the isolated point. On the three pieces:

- $\beta < 0$: $g(\beta) \leq -\frac{15}{4}$ by part (i).
- $0 < \beta \leq \frac{1}{4}$: g is increasing, so $g(\beta) \leq g(\frac{1}{4}) = \frac{1}{4} + 12 - 16 = -\frac{15}{4}$.
- $\beta = 1$: $g(1) = 3$.

So the greatest value of $\frac{1}{u} + \frac{1}{v} + uv$ is $\boxed{3}$, attained when $\beta = 1$, $\alpha = -2$, that is $u = v = 1$. (Indeed $1 + 1 + 1 = 3$ for both expressions.) Without that double root, nothing larger than $-\frac{15}{4}$ would be possible.

Checks

- Put $u = v = 1$ into both expressions: each is 3, matching part (iv). Put $u = v = -1$: each is -1, the equality case in part (iii).
- Try $\beta = \frac{1}{4}$ in part (iv): $\alpha = 1$, and $x^2 + x + \frac{1}{4} = (x + \frac{1}{2})^2$, so $u = v = -\frac{1}{2}$. Then $u + v + \frac{1}{uv} = -1 + 4 = 3$ and $\frac{1}{u} + \frac{1}{v} + uv = -4 + \frac{1}{4} = -\frac{15}{4}$, as claimed.

- A computer search over thousands of random β giving real roots found nothing above -1 in part (iii), and nothing above -3.75 in part (iv) apart from $\beta = 1$.

Common slips

- Forgetting the discriminant condition, and concluding in part (iii) that there is no upper bound.
- Solving $(\beta - 1)^2(4\beta - 1) \leq 0$ as just $\beta \leq \frac{1}{4}$ and missing $\beta = 1$, which gives the answer $-\frac{15}{4}$ instead of 3.
- Multiplying the inequality $\alpha^2 \geq 4\beta$ by β (which may be negative) rather than by β^2.

Beyond STEP

This is a small constrained optimisation problem: we maximise a function over the set of coefficients that give real roots, and that set can include isolated points as well as intervals. The maximum of a function on a set need not sit at a stationary point or at the end of an interval, which is why careful descriptions of the feasible set matter so much in analysis and optimisation. Abbott's *Understanding Analysis* gives a careful treatment of extreme values of continuous functions on general sets.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.